

Unit 101 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 97.4 ac, BAF 20 variable-radius cruise on a 300 ft grid, 45 plots, 478 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	212 sq ft/ac (SE 1.9 %)	Type	Sierran mixed conifer
Trees per acre	123	CWHR size / density	4 / D (cover 83 %)
QMD	17.8 in	SDI	288 (51 % of max 567)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	203 sq ft/ac 19.8 in	BA QMD	9 sq ft/ac 7.6 in
TPA	95	TPA	28
Volume	29.8 MBF/ac (5943 cu ft)	Volume	2.9 green tons/ac (130 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~146 sq ft/ac BA	Sampling error	3.9 % at 95 % (t = 2.02, n = 45)
Species (BA)	PP 70, WF 60, IC 32, DF 29, SP 21	Stratum standard	40 % - MEETS; 45 plots vs 20-plot practice: ok

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1). Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables exclude records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: [cruise_data.xlsx](#), Standards and assumptions.

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	8	15.6	3.6	51	0.3
8-11.9	34	27.7	15.1	241	1.2
12-15.9	74	30.7	32.9	670	3.4
16-19.9	82	21.2	36.4	870	4.4
20-23.9	68	11.7	30.2	831	4.2
24-27.9	50	6.0	22.2	686	3.4
28-31.9	52	4.8	23.1	782	3.9
32-35.9	33	2.3	14.7	534	2.7
36-39.9	25	1.4	11.1	439	2.2
40-43.9	18	0.9	8.0	310	1.6
44-47.9	10	0.4	4.4	173	0.9
48-51.9	3	0.1	1.3	59	0.3
52-55.9	9	0.3	4.0	189	0.9
56-59.9	4	0.1	1.8	88	0.4
60-63.9	7	0.2	3.1	151	0.8
All	477	123.4	212.4	6073	30.5

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
PP	158	45.2	70.2	16.9	2158	10.8
WF	134	31.5	59.6	18.6	1701	8.5
IC	71	19.1	31.6	17.4	768	3.9
DF	66	14.3	29.3	19.4	849	4.3
SP	47	13.3	20.9	17.0	574	2.9
PPP	1	0.1	0.4	35.1	23	0.1
All	477	123.4	212.4	17.8	6073	30.5

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 101: plots on the 300 ft grid (orange = QA flag)

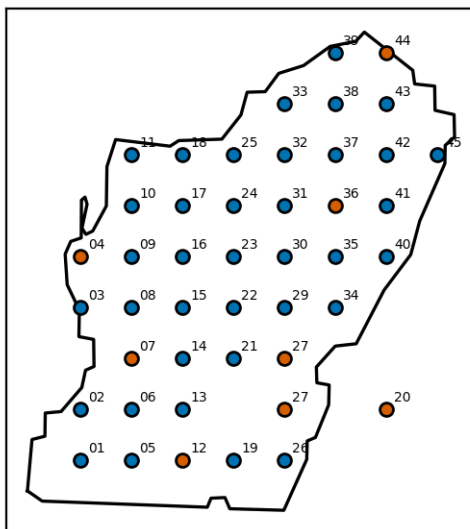


Figure 1. Plot locations with QA status.

QA findings: 7 item(s) need crew follow-up

Plot	Finding
101-04	tree 1: unknown species code 'PPP'
101-07	tree 9: height 32 ft implausible for DBH 27.8
101-12	tree 1: DBH 999.0 out of range
101-20	plot location falls outside its unit boundary (GPS or unit number error)
101-27	duplicate plot id recorded 2 times
101-36	tree 2: height missing
101-44	tree 1: height 15 ft implausible for DBH 34.0

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
101-01	12	240	11	ok
101-02	10	200	14	ok
101-03	6	120	28	ok
101-04	9	180	25	FLAG
101-05	9	180	16	ok
101-06	12	240	14	ok
101-07	13	260	9	FLAG
101-08	11	220	10	ok
101-09	11	220	19	ok
101-10	12	240	25	ok
101-11	9	180	24	ok
101-12	10	200	8	FLAG
101-13	11	220	5	ok
101-14	10	200	7	ok
101-15	10	200	3	ok
101-16	11	220	6	ok
101-17	11	220	4	ok
101-18	10	200	10	ok
101-19	10	200	8	ok
101-20	11	220	9	FLAG
101-21	11	220	10	ok
101-22	10	200	5	ok
101-23	11	220	3	ok
101-24	11	220	5	ok
101-25	11	220	6	ok
101-26	15	300	15	ok
101-27	11	220	9	FLAG
101-27	9	180	17	FLAG
101-29	12	240	8	ok
101-30	11	220	3	ok
101-31	11	220	3	ok
101-32	11	220	4	ok
101-33	9	180	7	ok
101-34	9	180	9	ok
101-35	10	200	2	ok
101-36	12	240	3	FLAG
101-37	11	220	6	ok
101-38	11	220	14	ok
101-39	10	200	4	ok
101-40	12	240	1	ok
101-41	12	240	2	ok
101-42	10	200	4	ok
101-43	9	180	8	ok
101-44	11	220	6	FLAG
101-45	10	200	10	ok